

JavaScript JSON & localStorage — Complete Answers

Answers for Tasks 1–15. Duplicate questions are answered once with the same solution. The browser examples can be copied into an HTML file and opened in a browser.

1. Flipkart-style product → JSON string

```
const product = {
  name: "boAt Rockerz 450",
  price: 1499,
  rating: 4.3
};

const jsonString = JSON.stringify(product);
console.log(jsonString);
```

Explanation: `JSON.stringify()` converts the JavaScript object into a JSON string.

2. Flipkart-style product → JSON string

```
const product = {
  name: "Samsung Galaxy M14",
  price: 12999,
  rating: 4.4
};

const jsonString = JSON.stringify(product);
console.log(jsonString);
```

Explanation: This is the same task as Task 1, using a different example product.

3. Top 3 food delivery apps → JSON string

```
const foodDeliveryApps = ["Zomato", "Swiggy", "Domino's"];

const jsonString = JSON.stringify(foodDeliveryApps);
console.log(jsonString);
```

Explanation: An array is also converted to a JSON string with `JSON.stringify()`.

4. JSON string of YouTube channels → array of names

```
function getChannelNames(jsonString) {
  const channels = JSON.parse(jsonString);
  return channels.map(channel => channel.name);
}

const jsonData = `[
  {"name": "CodeWithHarry"},
  {"name": "Apna College"},
  {"name": "freeCodeCamp"}
]`;

console.log(getChannelNames(jsonData));
```

Explanation: The function parses the JSON and maps each object to its name.

5. JSON string of YouTube channels → array of names

```
function getChannelNames(jsonString) {
  const channels = JSON.parse(jsonString);
```

```

    return channels.map(channel => channel.name);
  }

  const jsonData = `[
    {"name": "MrBeast"},
    {"name": "Kurzgesagt"},
    {"name": "Fireship"}
  ]`;

  console.log(getChannelNames(jsonData));

```

Explanation: Same requirement as Task 4, with another example list.

6. saveRecentSearch(query)

```

function saveRecentSearch(query) {
  const searches = JSON.parse(
    localStorage.getItem("recentSearches") || "[]"
  );

  searches.push(query);

  localStorage.setItem(
    "recentSearches",
    JSON.stringify(searches)
  );
}

saveRecentSearch("Biryani");
saveRecentSearch("Sneakers");

```

Explanation: The existing searches are read first, the new query is added, and the complete array is saved as JSON.

7 & 8. Favorite Zomato restaurants — save/load with localStorage

```

<!DOCTYPE html>
<html>
<head>
  <title>Favorite Restaurants</title>
</head>
<body>
  <h2>My Top 3 Zomato Restaurants</h2>

  <input id="restaurantInput"
    placeholder="Enter restaurant name">
  <button onclick="addRestaurant()">Add</button>
  <button onclick="clearFavorites()">Clear Favorites</button>

  <ul id="restaurantList"></ul>

  <script>
    let restaurants =
      JSON.parse(localStorage.getItem("favoriteRestaurants")) || [];

    function displayRestaurants() {
      const list = document.getElementById("restaurantList");
      list.innerHTML = "";

      restaurants.forEach(restaurant => {
        const li = document.createElement("li");
        li.textContent = restaurant;
        list.appendChild(li);
      });
    }

    function addRestaurant() {
      const input = document.getElementById("restaurantInput");

```

```

    const restaurant = input.value.trim();

    if (!restaurant) return;

    if (restaurants.length >= 3) {
        alert("You can add only 3 restaurants.");
        return;
    }

    restaurants.push(restaurant);

    localStorage.setItem(
        "favoriteRestaurants",
        JSON.stringify(restaurants)
    );

    input.value = "";
    displayRestaurants();
}

function clearFavorites() {
    localStorage.removeItem("favoriteRestaurants");
    restaurants = [];
    displayRestaurants();
}

displayRestaurants();
</script>
</body>
</html>

```

Explanation: This single page handles both adding and saving the top 3 restaurants and automatically loading them after a page reload.

9. getRecentSearches()

```

function getRecentSearches() {
    const saved = localStorage.getItem("recentSearches");

    if (saved === null) {
        return [];
    }

    return JSON.parse(saved);
}

console.log(getRecentSearches());

```

Explanation: The null check prevents `JSON.parse(null)` from being used when nothing has been saved.

10 & 11. Clear Favorites button

```

function clearFavorites() {
    localStorage.removeItem("favoriteRestaurants");
    restaurants = [];
    displayRestaurants();
}

```

Explanation: This removes the saved key and immediately refreshes the UI. Tasks 10 and 11 are duplicates.

12. Flipkart-style wishlist with localStorage

```

const wishlist = [
    { name: "Wireless Headphones", price: 1999 },
    { name: "Smart Watch", price: 2499 },
    { name: "Running Shoes", price: 1799 }
]

```

```

];

localStorage.setItem("wishlist", JSON.stringify(wishlist));

// Read the wishlist after a page reload
const savedWishlist =
  JSON.parse(localStorage.getItem("wishlist")) || [];

console.log(savedWishlist);

// Display the wishlist in a page
const list = document.getElementById("wishlist");
savedWishlist.forEach(product => {
  const li = document.createElement("li");
  li.textContent = `${product.name} - ₹${product.price}`;
  list.appendChild(li);
});

```

Explanation: Add to the HTML before running the display section.

13 & 14. Debug the movie localStorage code

```

// WRONG:
// localStorage.setItem("movies", movies);

let movies = ["Jawan", "Pathaan", "Dunki"];

// Correct: convert the array to JSON before saving.
localStorage.setItem("movies", JSON.stringify(movies));

let saved = localStorage.getItem("movies");
let arr = JSON.parse(saved);

console.log(arr);

```

Explanation: localStorage stores values as strings. Saving an array directly turns it into a comma-separated string such as "Jawan,Pathaan,Dunki", which is not valid JSON. Use JSON.stringify() when saving and JSON.parse() when reading. Tasks 13 and 14 are duplicates.

15. Clear Wishlist button

```

function clearWishlist() {
  localStorage.removeItem("wishlist");
  displayWishlist();
}

function displayWishlist() {
  const list = document.getElementById("wishlist");
  list.innerHTML = "";

  const savedWishlist =
    JSON.parse(localStorage.getItem("wishlist")) || [];

  savedWishlist.forEach(product => {
    const li = document.createElement("li");
    li.textContent = `${product.name} - ₹${product.price}`;
    list.appendChild(li);
  });
}

```

Explanation: HTML button: Clear Wishlist. removeItem() deletes the saved wishlist, and displayWishlist() immediately updates the page.

Quick JSON + localStorage rules

Object/array → **JSON**: `JSON.stringify(value)`

JSON → **JavaScript object/array**: `JSON.parse(jsonString)`

Save: `localStorage.setItem(key, JSON.stringify(value))`

Read: `JSON.parse(localStorage.getItem(key))`

Delete: `localStorage.removeItem(key)`

Important: `localStorage` persists across page reloads, but it stores data as strings. Therefore, arrays and objects should normally be serialized with `JSON.stringify()` before saving and deserialized with `JSON.parse()` after reading.